

The boundary-spanning behavior of nurses: The role of support and affective organizational commitment

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Aim: The aim of this study was to examine the relationship between organizational, supervisor, and coworker support, as perceived by registered nurses and their boundary-spanning behaviors. Furthermore, this article examines the mediating role of the affective organizational commitment of nurses in these relationships.

Background: Registered nurses play a key role in hospitals, as they have an important impact on the quality of the services delivered. For nurses to perform at their best, they need organizational, leader, and coworker support. To date, few studies have explored the link between nurses' perceived support, affective organizational commitment, and boundary-spanning behaviors.

Methods: This cross-sectional research used a questionnaire survey to explore the hypothesized relationships in a sample of 273 nurses from a hospital in Belgium. Structural equation modeling was used for statistical analysis of the mediation model.

Results: One hundred forty-seven (53.5%) nurses responded to the survey. Perceived support from the organization, supervisors, and coworkers positively influences nurses' boundary-spanning behaviors. Affective organizational commitment was found to mediate the positive relationship between perceived organization support, perceived coworker support, and boundary-spanning behaviors. Perceived supervisor support and boundary-spanning behaviors showed a direct relationship not mediated by affective organizational commitment.

Conclusions: Perceived support has an important influence on the boundary-spanning behavior of nurses. This study emphasizes the importance on how support exerts an influence on boundary-spanning behavior and

Key words: affective organizational commitment, boundary-spanning behavior, perceived support, social exchange theory

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underscores the importance of affective organizational commitment. Health care organizations, supervisors, and coworkers are essential in fostering boundary-spanning behaviors of nurses, both directly and through the development of affective organizational commitment. These actors should therefore be aware of the way they behave and the implications their behavior may have.

Registered nurses play a central important role in health care organizations. They interact closely with patients and play a key front office role (Verleye, Gemmel, & Rangarajan, 2016). Nurses' engagement and customer-oriented behavior are critical in improving patients' well-being and quality of care (Trybou & Gemmel, 2016). It is crucial for organizations to support and motivate nurses to perform boundary-spanning behaviors. This behavior can be seen as a type of extra role behavior that goes beyond the formal requirements of the job. With high levels of extra role behavior, nurses tend to go the extra mile, even in less significant circumstances. Nurses become more proactive and become more engaged in organizational activities (Trybou, De Pourcq, Paeshuyse, & Gemmel, 2014). Here, we concentrate on three distinct forms of nurses' boundary-spanning behavior: service delivery (similar to key quality service elements), external representation (as an external advocate of the company), and internal influence (as an internal champion for delivering service excellence) (Bettencourt, Brown, & Mackenzie, 2005; Trybou & Gemmel, 2016).

Previous studies have suggested that extra role behavior is influenced by the organizational support perceived by nurses (Chang, 2014). As nurses develop relationships with their organization and colleagues, multiple sources of support are created. The quality of these relationships has a significant impact on the attitudes of nurses, their career choices, and performances (Rodwell, McWilliams, & Gulyan, 2017). However, previous research also shows that, as nurse leaders build intense relationships with the nurses they supervise, head nurses can have an impact on these processes and that these processes are more complex than originally conceived (Trybou & Gemmel, 2016).

The aim of this research was to study the relationship between three sources of support (from the organization, from the supervisor, and from the colleagues) and registered nurses' boundary-spanning behaviors (service delivery, internal influence, and external representation). In addition, affective organizational commitment—a person's emotional attachment to, identification with, and involvement in an organization (Allen & Meyer, 1990)—is studied as a potential mediator of this relationship, as it has been shown that nurses who work in more supportive work environments tend to be more committed (Sadatsafavi, Walewski, & Shepley, 2015).

By exploring these relationships, this article aims to find further evidence that perceived support has an impact on the degree of boundary-spanning behavior of nurses. In

doing so, we contribute to the literature. First, there have been few studies that have focused on the perceived support and extra role behavior of nurses (Trybou et al., 2014). Second, affective commitment has been linked to favorable organizational behaviors; we investigate whether it is potentially related to boundary-spanning behaviors. Third, this study makes a contribution on account of the sample it employs: examining health care workers' mechanisms of motivation and especially those of nurses has high priority in current research agendas (Gellatly, Cowden, & Cummings, 2014).

Framework

Customer-Oriented Boundary-Spanning Behaviors

The service management and boundary-spanning literature both suggest three key dimensions of customer-linking behaviors (Bettencourt et al., 2005): (a) external representation, being vocal advocates to outsiders of the organizations image, goods, and services; (b) internal influence, taking individual initiative in communications to the firm and coworkers to improve service delivery by the organization, coworkers, and oneself; and (c) service delivery, serving customers in a conscientious, responsive, flexible, and courteous manner (Bettencourt et al., 2005). Nursing professionals, in their role as frontline service employees, show these boundary-spanning behaviors, including role-prescribed and extra role customer services (Hsu, Chang, Huang, & Chiang, 2011); they contribute to the quality of the services provided and the image and legitimacy of the organization (Mollahosseini, Kamama, & Mirhosseini Sarat Layegh, 2012). The construct customer-oriented boundary-spanning behavior with the three dimensions has been found valuable in understanding boundary-spanning behavior in prior research (Bettencourt et al., 2005; Trybou & Gemmel, 2016).

Social Exchange Theory

This article is guided by social exchange theory as a framework to explain the behavior of nurses in relation to the perceived support they receive.

According to social exchange theory, social exchange involves a series of interactions generating obligations

(Emerson, 1976). Social exchanges evolve over time into mutual commitments based on rules of exchange. To ensure that the exchange relationship can evolve into a relation of mutual commitment, trust, and loyalty, all actors apply “a set of rules of exchange” (Cropanzano & Mitchell, 2005). One of these rules is “the norm of reciprocity,” whereby actions by one party lead to a moral debt on the part of another. The norm of reciprocity does not only exist between people but also between people and organizations. This is because employees assign humanlike attributions to the organizations they deal with (Levinson, 1965). The noncompliance of these rules of game can, however, lead to termination of the exchange relationship. To assess the value of the exchange relationship, there are different tools. One of these is the measure of perceived organizational support (POS). The organizational support theory (Eisenberger, Huntington, Hutchison, & Sowa, 1986) assumes that employees make assumptions about their organization and the measure to which it values their contributions and well-being. The purpose of these general assumptions is to create an overview of the measure into which the organization rewards job efforts and to estimate if the organization fulfils the socioemotional needs of the employee. This enables the employee to assess the measure of support the employee can expect when this is needed to conduct his or her job, to cope with stressful situations, and to judge if an organization wants to continue the social exchange relationship. The higher the measure of POS, the higher the moral debt will be for a similar action by the organization. This assessment is not only conducted for the organization but also for the coworkers (perceived coworker support [PCS]). PCS, as a construct, can be interpreted as “the extent to which employees believe their coworkers are willing to provide them with work related assistance to aid in the execution of their service-based duties” (Susskind, Kackmar, & Borchgrevink, 2003, p. 181). Higher measures of perceived support leads to favorable behavior, which includes but is not limited to an increase in affective commitment (Kim, Eisenberger, & Baik, 2016), work engagement (e.g., Ancarani, Di Mauro, & Giammanco, 2017a), in-role (e.g., Ancarani, Di Mauro, Giammanco, & Giammanco, 2017b) and extra role behavior (e.g., Gupta, Agarwal, & Khatri, 2016), and customer behavior (e.g., Susskind et al., 2003). Social exchange theory states that nurses tend to reciprocate beneficial treatment they experience with positive work related behavior.

Nurses' Perceived Support

Previous studies suggest that extra role behavior is influenced by the support perceived by nurses (Trybou et al., 2014). As nurses develop relationships with their organization, supervisor, and colleagues, multiple sources for support are created. The quality of these relationships and, therefore, the measure of support have a significant impact

on the attitudes of nurses and their career choices and performances (Ancarina et al., 2017b).

Nurses engage in visible and invisible social exchanges with other professionals and their organization (Trybou et al., 2014). Exchange relationships exist between the nursing professional, his or her coworkers, the head nurse, and the hospital as his or her employer (Anacari et al., 2017b). Chang (2014) studied the influence of organizational support on nurses' organizational citizenship behaviors. Nurses who feel that their organization is treating them impartially and fairly and providing supportive emotional attachment tend to reciprocate with behaviors that benefit the organization. The subjective perceptions of nursing staff do play a crucial role in the exchange relationship with the organization; the greater the level of organizational justice perceived by the nursing staff, the more they will support the organization and perceive and exhibit citizenship behaviors, such as altruism and dedication to their work (a similar construct to extra role behavior), thus setting up a positive relationship (Chang, 2014). Chenevert, Vandenberghe, and Tremblay (2015) also found that the perception of organizational support exerts a positive influence on such organizational citizenship behaviors. These studies demonstrated positive associations between behavior and organizational support. Greater perception of organizational support by nurses thus tends to lead to more customer-oriented boundary-spanning behaviors. As such, the following hypothesis was proposed:

Hypothesis 1. POS is positively associated with boundary-spanning behaviors.

The assessment of exchanges by nurses considers not only the organization but also the coworkers. PCS, as a construct, can be described as the belief of employees that coworkers are willing to assist and aid with service-based duties (Susskind et al., 2003). Higher levels of PCS may lead to positive organizational behavior, including an increase in job performance and satisfaction (Shin, Hur, & Choi, in press), affective organizational commitment, organizational citizenship behavior (Chenevert et al., 2015), and customer-oriented behavior (Susskind et al., 2003). We thus propose the following hypothesis:

Hypothesis 2. PCS is positively associated with boundary-spanning behaviors.

In the same way that employees form general perceptions of how their organization values them, they also tend to develop global views of how much their supervisors value their contributions and care about their well-being (perceived supervisor support [PSS]; Kottke & Sharafinski, 1988). In their groundwork, Levinson (1965) and Eisenberger et al. (1986) posed that supervisors act as agents of the organization and are responsible for evaluating and directing

the performance of employees. Employees often see their supervisor's unfavorable or favorable attitude to them as being indicative of the support provided for them by the organization. However, at the same time, research stated that employees are capable of distinguishing their relations with the organizational entity and its leaders from their relations with their direct supervisors (Dirks & Ferrin, 2002). Employees become attached independently to each focus as a result of making this distinction (Vandenberghe, Bentein, & Stinglhamber, 2004). Outside the nursing setting, various effects of PSS have been found, such as its supportive relationship with the supervisor increasing job satisfaction (Gok, Karatuna, & Osdemir Karaca, 2015) and its relation to turnover intentions and work engagement (Ancarani, Di Mauro, Giammanco, & Giammanco, 2017b). As such, it is interesting to study the relationship between PSS and customer-oriented boundary-spanning behaviors of nurses. We thus hypothesize that nurse supervisors who do support their employees will explain significant variance in outcome.

Hypothesis 3. PSS is positively associated with boundary-spanning behaviors.

Mediating Role of Affective Organizational Commitment

Many researchers have studied the impact of perceived support and employees attitudes. These attitudes are both favorable to employees (job satisfaction and psychological well-being) and the organization (affective commitment, turnover intentions, work performance, and work engagement). Affective organizational commitment is one of the most important organizational attitudes (Ancarani et al., 2017a; Ancarani et al., 2017b; Rhoades & Eisenberger, 2002). It was defined by Allen and Meyer (1990) as a person's emotional attachment to, identification with, and involvement in a particular organization. Individuals who are committed affectively to the organization tend to prefer to remain in that organization, usually share the goals of the organization, are satisfied with their work, and feel they fit in. Employees with affective commitment also feel valued, are positive assets for the organization, and act as organizational ambassadors. Blau (1968) argued that social exchange behavior depends on an individual's evaluation of intrinsic and extrinsic rewards. Affective organizational commitment can be directed toward different targets or foci including the supervisor (Landry & Vandenberghe, 2009). This clarifies why higher levels of affective commitment lead to more positive organizational behavior (Allen & Meyer, 1990).

However, social exchanges and reciprocity are more complex than originally conceived. Moreover, it can be argued that perceived support first leads to affective reactions and, thereafter, to boundary-spanning behavior (Bishop, Scott,

& Burroughs 2000). The effect on boundary-spanning behavior therefore depends on an interpretation process. Nurses engage in cognitive sense-making, with the aim of producing meaning from the cognitions of support. This offers a way to understand how nurses react to the level of experienced support of the organization, colleagues, and head nurse. Therefore, affective organizational commitment is included as a potential mediator

Hypothesis 4. Affective organizational commitment mediates the relationship between perceived support (POS, PSS, and PCS) and boundary-spanning behaviors.

Given the above, in this study, support perceptions (POS, PSS, and PCS) serve as independent variables, customer-oriented boundary-spanning behavior is taken as the dependent variable, whereas affective organizational commitment serves as the mediating variable (see Figure 1).

Methods

Procedure and Participants

A cross-sectional study design was used to investigate the link between perceived support, affective commitment, and boundary-spanning behaviors. The paper version questionnaire was made available to all 273 nurses at one hospital in the northern region of Belgium. The hospital has about 400 beds and 1,200 employees. Data collection was performed in March and April 2015.

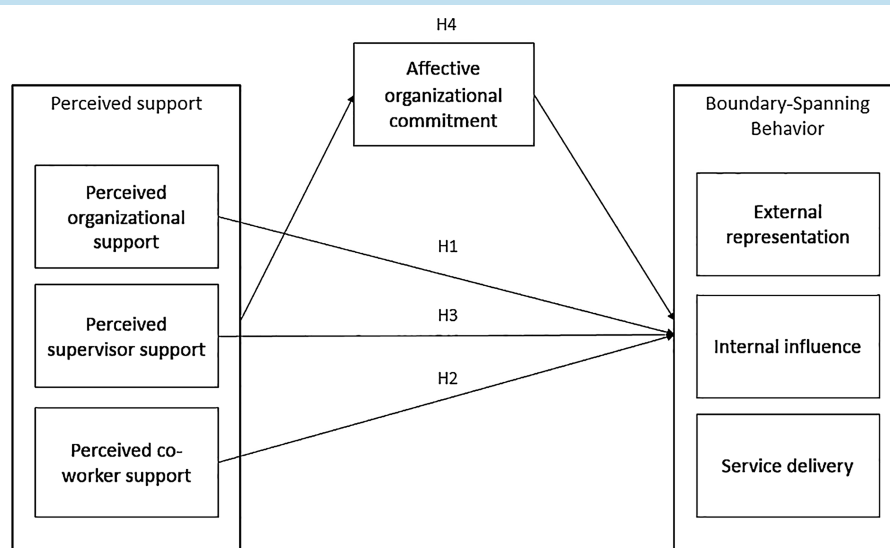
The sample consisted of 147 valid responses, yielding a response rate of 53.5%. Of the 147 valid respondents, 86% were female and 14% were male. The average age of respondents was 38 years (23–64 years, 11.82%). Forty-two percent of the participants had job tenure of less than 5 years, 22% between 5 and 10 years, 11% between 11 and 20 years, and 25% over 20 years. Fifty-six percent of the participants worked full time.

Ethical Consideration

The university-affiliated institutional review board approved the study protocol. All participants agreed to participate through informed consent. This study has been conducted according to the guidelines of Good Clinical Practice and the Declaration of Helsinki for Medical Research Involving Human Subjects.

Measures

All measures were administered in Flemish. Specifically, all original scales were translated in Flemish using the standard back-translation technique. A demographic

Figure 1**The hypothesized research model**

questionnaire was incorporated to obtain descriptive information. Individuals' gender (Cameron & Nadler, 2013), organizational tenure (Dulac, Coyle-Shapiro, Henderson, & Wayne, 2008), and percentage of full-time employment (workload; Clinebell & Clinebell, 2007) were included to rule out potential alternative explanations for our findings. Besides questions regarding demographics, the questionnaire involved seven different constructs. Each construct was constructed out of previously validated instruments found in the literature. Construct validity and reliability were examined in previous studies and were found to be adequate. The reliability indices can be found in the result section (Table 1). All the free factor loadings are statistically significant. Consequently, the construct validity of the seven latent variables appears to be at an acceptable level.

Perceived organizational support. Previous research provides evidence of the internal reliability and unidimensionality of the POS measure (Eisenberg et al., 1986). An eight-item short form by Eisenberger et al. (1986) was used ($\alpha = .715$ in our study sample). Responses were provided on a 5-point Likert scale, with anchors ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). One of the eight questions for POS is "My organization values my contribution to its well-being."

Perceived coworker support. Following Hayton, Camabuci, and Eisenberger (2012), PCS was measured by the same eight questions but enquiring about the "coworker" instead

of the "organization" ($\alpha = .892$ in the current sample). These six items included items like "my coworkers care about my well-being." Responses were given on a 5-point Likert scale, and items were intended to draw out the respondents' beliefs about their coworkers' general attitudes to them. In previous studies, this scale has shown very satisfactory reliability and validity.

Perceived supervisor support. In order to assess the perceptions of employees that their supervisors care about their well-being and value their contributions, we used the POS scale in its adaptation by Rhoades and Eisenberger (2002); this demonstrated high levels of internal reliability ($\alpha = .89$); for other items, the word "organization" was replaced by "supervisor" ($\alpha = .900$ in the current sample). We used a 5-point Likert scale.

Boundary-spanning behavior. The questionnaire of Bettencourt and Brown (2003) was used to measure boundary-spanning behavior. All three outcome variables were measured with three items each. A sample question for external representation is "I tell outsiders this is a great place to work" (Cronbach's $\alpha = .90$, Bettencourt & Brown, 2003; $\alpha = .81$ in the current sample). A sample question for internal influence is "I give constructive proposals for improving the services" (Cronbach's $\alpha = .94$, Bettencourt & Brown, 2003; $\alpha = .87$ in the current sample). An example question for service delivery is "I am courteous and respectful towards patients, no matter the circumstances" (Cronbach's $\alpha = .91$, Bettencourt & Brown, 2003; $\alpha = .76$

Table 1
Cronbach’s alpha and correlations

Variable	Mean (min–max; SD)	Cronbach’s α	1	2	3	4	5	6	7	8	9
1. Gender			—								
2. Organizational tenure			–.095	—							
3. Workload			.113	–.507**	—						
4. Perceived organizational support	3.38 (1.75–4.5; 0.507)	.715	–.080	–.028	–.002	—					
5. Perceived coworker support	4.08 (2.13–5; 0.577)	.892	–.156	.025	–.020	.451**	—				
6. Perceived Supervisor support	4.06 (2–5; 0.664)	.900	.005	–.011	–.111	.447**	–.258**	—			
7. Affective organizational commitment	3.33 (1.75–5; 0.582)	.743	–.176*	.224*	–.173*	.521**	.586**	.300**	—		
8. External representation	3.89 (2–5; 0.676)	.812	–.075	–.034	–.040	.439**	.467**	.464**	.421**	—	
9. Internal influence	3.55 (1–5; 0.675)	.865	–.005	.232*	–.137	.284**	.351**	.244**	.390**	.435**	—
10. Service delivery	4.17 3–5; 0.502)	.762	–.077	.128	–.167	.074	.103	.080	.130	.345**	.358**

Note. $N = 151$.
* $p < .05$. ** $p \leq .001$.

in the current sample). Responses were made on a scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*).

Affective organizational commitment. We used five items from Allen and Meyer’s Affective Commitment Scale (Allen & Meyer, 1990) to assess affective organizational commitment; a sample question is “I think I could easily become as attached to another organization as I am to this one.” Research indicated that this scale functions as a single high-reliability factor (Allen & Meyer, 1990). The internal reliability proved acceptable ($\alpha = .74$).

Data Analysis

SPSS software (Version 24, IBM, Chicago, IL) was used for sample analysis and descriptive statistics. The sample and study variables underwent descriptive statistical analysis. Cronbach’s alpha reliability scores were calculated to test the internal consistency of the scales. The scales demonstrated adequate internal consistency (see Table 1).
We used structural equation modeling (SEM) to test the hypotheses. LISREL 9.3 was used for parameter estimation and SEM evaluation. SEM is a flexible approach to modeling relationships between variables. It allows the simultaneous

estimation of the relations between dependent and independent variables (the structural model) and between latent and observed variables (the measurement model). Because it allows estimation of a model in its entirety, it is a more reliable approach than other methods such as hierarchical regression or analysis of variance (Hoyle & Smith, 1994). Specifically for mediation analysis, SEM allows to test the direct and indirect effects simultaneously, therefore allowing a more robust test of mediation than the traditional procedure.
 p Values are reported as two-tailed with a significance level (α) of .05.

Model Fit

The fit indices indicate a reasonable to good fit of the model. $\chi^2/df = 2.66$ (norm < 3), RMSEA = 0.1 (norm < 0.1), and SRMR = 0.09 (norm < 0.1). The model showed no problematic levels of multicollinearity. The condition number equaled 22, which is well below the norm of 30. The R^2 values for the paths leading to the dependent variables was .53 for affective organizational commitment, .24 for internal influence, .47 for external representation, and .19 for service delivery.

Results

Descriptive Statistics and Intercorrelations

As a first step, all variable correlations were calculated (Table 1). There is a significant relationship between job tenure and workload time ($r = -.51, p \leq .001$), between job tenure and affective organizational commitment ($r = .22, p = .008$), and between job tenure and internal influence ($r = .23, p < .005$). These correlations imply that registered nurses with longer tenure have a tendency to work fewer hours per week and to display more affective commitment and internal influence. Registered nurses who work more hours per week show lower levels of affective organizational commitment ($r = -.17, p = .046$). Male nurses show higher affective organizational commitment ($r = .18, p < .05$).

There were significant correlations between POS and PCS ($r = .45, p \leq .001$), PSS ($r = .45, p \leq .001$), affective commitment ($r = .52, p \leq .001$), external representation ($r = .44, p \leq .001$), and internal influence ($r = .28, p \leq .001$). Other similar results were found between PCS and PSS ($r = -.26, p \leq .001$), affective organizational commitment ($r = .59, p \leq .001$), external representation ($r = .47, p < .001$), and internal influence ($r = .35, p \leq .001$). PSS correlated with affective organizational commitment ($r =$

$.30, p \leq .001$), external representation ($r = .46, p < .001$), and internal influence ($r = .22, p = .004$).

All three constructs for boundary-spanning behavior showed significant positive correlations at a level of $p \leq .001$ (see Table 1). This implies that registered nurses who display one type of boundary-spanning behavior are more likely to also demonstrate other types.

Main Analyses

We performed mediation analysis following the guidelines for mediation analysis in SEM from Zhao, Lynch, and Chen (2010). We estimated a model that contains both the direct effects (between the dependent and independent variables) and the indirect effects, which are mediated by the variable affective organizational commitment. The results of the analysis can be found in Table 2 and Figure 2.

Hypothesis 1. POS is positively associated with boundary-spanning behaviors.

Hypothesis 2. PCS is positively associated with boundary-spanning behaviors.

Hypothesis 3. PSS is positively associated with boundary-spanning behaviors.

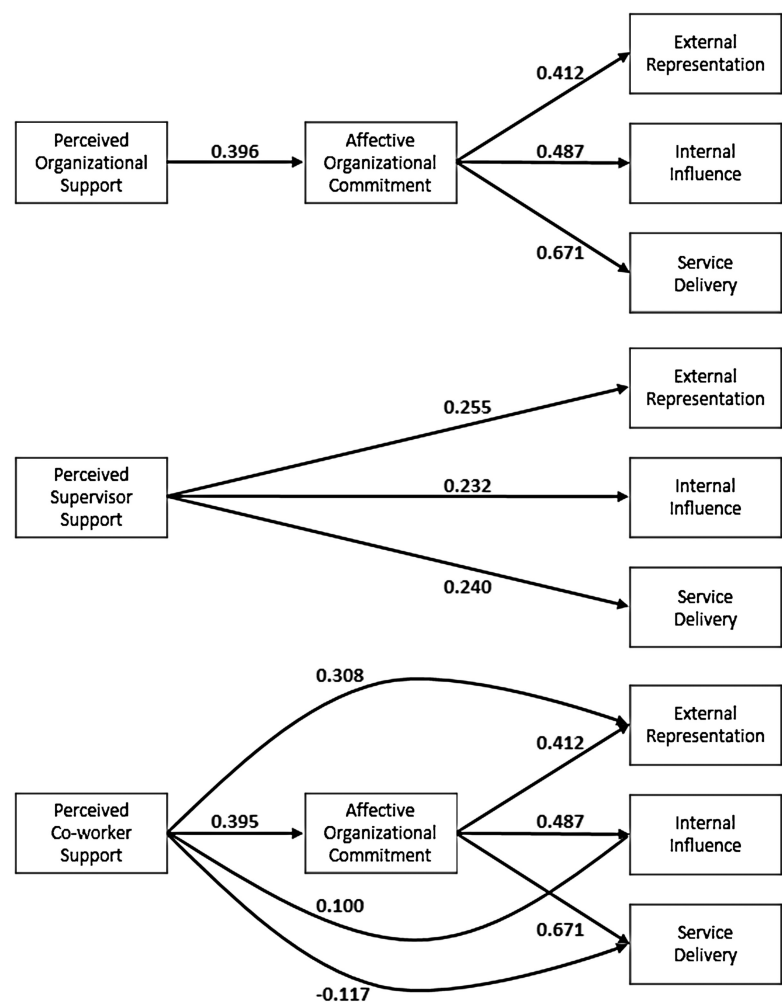
Table 2

Main analysis

Effect	Direct effect		Indirect effect		Acceptance
	β	p	r	p	
POS $\Rightarrow$ External	-.061	not sig.	.163	.024	Full mediation
POS $\Rightarrow$ Internal	-.082	not sig.	.193	.026	Full mediation
POS $\Rightarrow$ Service	-.468	not sig.	.266	.013	Full mediation
POS $\Rightarrow$ AOC	.396	<.001	—	—	
PSS $\Rightarrow$ External	.255	<.001	.045	not sig.	No mediation
PSS $\Rightarrow$ Internal	.232	<.001	.053	not sig.	No mediation
PSS $\Rightarrow$ Service	.240	<.001	.073	not sig.	No mediation
PSS $\Rightarrow$ AOC	.109	not sig.	—	—	
PCS $\Rightarrow$ External	.308	<.001	.163	.024	Partial mediation
PCS $\Rightarrow$ Internal	.100	.012	.192	.014	Partial mediation
PCS $\Rightarrow$ Service	-.117	<.001	.265	.005	Partial mediation
PCS $\Rightarrow$ AOC	.395	<.001	—	—	
AOC $\Rightarrow$ External	.412	.002	—	—	
AOC $\Rightarrow$ Internal	.487	.002	—	—	
AOC $\Rightarrow$ Service	.671	<.001	—	—	

Note. For the measurement of the indirect effect, the variable AOC was used as a mediator. PSS = perceived supervisor support; POS = perceived organizational support; PCS = perceived coworker support; AOC = affective organizational support; Internal = internal influence; External = external representation; Service = service delivery.

Figure 2
The structural model



Arrows indicate significant relationships. Standardized betas are given for each relationship.

The first three hypothesis were accepted. There exists a relationship between the three different types of support and the three different types of boundary-spanning behaviors. The fourth hypothesis, “Affective organizational commitment mediates the relationship between perceived support (POS, PSS and PCS) and boundary-spanning behaviors,” could not be accepted. The relationship between PSS and boundary-spanning behaviors is not mediated by affective commitment. The relationship between POS and boundary-spanning behaviors, on the other hand, is fully mediated by affective organizational commitment. The relationship

between PCS and boundary-spanning behaviors is partially mediated by affective organizational commitment.

Discussion

Hospitals face an ever-competitive health care environment. Registered nurses play key front-office roles in hospitals and have an important impact on the quality of the services provided to patients (Galletta et al, 2013). For nurses to perform at their best, they need the necessary support from

the organization, colleagues, and leader. The estimates of the model have several implications regarding the hypotheses. First, there exists a relationship between the three different types of support and the three different types of boundary-spanning behaviors. This indicates that perceived support on the work floor, be it from coworkers, supervisors, or the organization, leads to nurses going the extra mile. Second, the paths leading from perceived support to boundary-spanning behaviors are different for the three distinct types of perceived support. The relationship between PSS and boundary-spanning behaviors is not mediated by affective organizational commitment. This suggests that the relation between PSS and boundary-spanning behaviors is either a direct relationship or is mediated by a different variable, not included in this model. Ng and Sorensen (2008) argued that it is possible that PSS has a stronger and different kind of relationship with work attitudes than POS and PCS. Support perceived from the direct supervisor may be seen as more valuable. Guiding employees is part of the job description of supervisors, so they may be more experienced at this skill. Supervisors are a more stable resource of support, which can be convenient to recall more supportive actions. Another possibility is that nurses do distinguish between their sources of support and direct their commitment directly to these entities (Vandenberghe et al., 2004) and that boundary-spanning behaviors are a result of reciprocation upon each of these relations separately. This could also explain the fact that affective commitment only partially mediated the effect of PCS on boundary-spanning behavior. However, because of limitations in their research, Vandenberghe et al. (2004) warn to be cautious about interpreting this causality.

In this study, the relationship between POS and boundary-spanning behaviors is fully mediated by affective organizational commitment. In other words, we do not observe a direct relationship between POS and boundary-spanning behaviors. Instead, POS is positively related to affective organizational commitment, and affective organizational commitment, in turn, is positively related to boundary-spanning behaviors. The relationship between PCS and boundary-spanning behaviors is partially mediated by affective organizational commitment. In other words, we observe a direct relationship between PCS and boundary-spanning behaviors, and we also observe an indirect relationship, which is mediated by affective organizational commitment. This reinforces previous findings supporting the conjecture that commitment arises directly in response to POS and PCS (Gupta et al., 2016) and explanatory mechanisms for the effects of POS and PCS on the boundary-spanning behavior of nurses. Nurses who receive support from the organization are more likely to identify and emotionally bond with it (Gupta et al., 2016).

It can be argued that the extent to which perceived support leads to affective reactions (and ultimately boundary-spanning behavior) depends on an interpretation process.

Similar to psychological contracts (Morrison & Robinson, 1997), with which it shares similar features like the reciprocity norm, it could be that nurses engage in cognitive sense-making with the aim of producing meaning from the cognitions of support. Recent literature supports this view as both constructs are more and more studied together (e.g., Ahmad & Zafar, 2018; Gupta et al., 2016). This offers a way to understand how nurses react to the level of experienced support of the organization, colleagues, and head nurse.

This article adds to the literature on commitment by extending knowledge on the relationship between affective organizational commitment and nurses' boundary-spanning behavior. Indeed, previous research mostly studied the effects of nurses on work-related outcomes (e.g., job performance) but did not pay as much attention to boundary-spanning behavior. This research therefore enriches understanding of the effects of affective organizational commitment on boundary-spanning behavior of nurses toward the customer.

The data collection was carried out in a single general hospital. This could lead to one-sided data as the perceptions and norms within the same organization may be similar among nursing professionals. Future studies should focus on multiple organizations. Because the research design is cross-sectional, statements about causality between perceived support and boundary-spanning behavior should be treated with care. A longitudinal study would be more valuable.

In their work environment, nursing professionals engage in different social exchange relationships. In this article, the relationship with the organization, direct supervisor, and direct colleagues was studied. However, other relationships do exist—for example, with physicians, other health care professionals, patient relatives, and others. It would therefore be valuable to study the impact of other relationships on registered nurses' organizational attitudes and behavior.

Practice Implications

Although not all our hypotheses were supported, the results show the importance of all three sources of support to registered nurses. The more nurses perceive that the organization supports them and cares about their well-being, the more they feel obligated to return the favor and build up a positive relationship with the organization. This will increase the levels of affective commitment, which in turn has a positive relationship with customer-oriented boundary-spanning behavior.

Support nurses receive from the supervisor is not mediated through affective commitment, suggesting the possibility of a direct relationship between the two constructs. This could implicate that supervisors have a direct influence on the level of boundary-spanning behaviors the nurses they lead demonstrate. Organizations should take the influence of the direct supervisor under consideration when trying

to enhance boundary-spanning behavior by front-office employees. Supervisors should be aware of the considerable part they play on nurses' extra role behavior and thus support and mentor them properly.

Support provided by coworkers has both a positive direct relationship with boundary-spanning behavior and an indirect relationship through affective commitment. The sense you can count on your colleagues and work together toward a shared goal will increase extra role behavior. Coworker support leading to affective organizational commitment could be explained as nurses see each coworker as an individual and coworkers as a group that is part of "the organization." As such, coworkers beyond all departments are also part of the organization; they develop feelings of affective commitment. Organizations should be aware that they also influence boundary-spanning behavior through coworkers. For managers, it is important to promote coherent teams with a positive atmosphere, which facilitates a collegial work environment to increase the support nurses perceive. It is important that health care managers acknowledge the impact that the organization, supervisors, and coworkers have on employees. The level of support nurses perceive will influence the extent to which boundary-spanning behaviors are demonstrated. Therefore, nursing managers could benefit from considering the outcomes of this study, as it may enhance managerial actions to promote extra role behavior.

Conclusion

This article found that POS, PSS, and PCS are positively related to boundary-spanning behavior. Affective organizational commitment mediates the relationship fully for POS and partially for PCS and boundary-spanning behavior of nurses. PSS was not found to be mediated by affective organizational commitment. These results demonstrate that the organization and coworkers play an important part in the affective organizational commitment of the nurses and that both organization staff and coworkers should therefore be aware of the way they behave and the implications that their behavior may have. It also underlines the important role the direct supervisor plays toward the boundary-spanning behavior of nurses.

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